

ECE 443/518 – Computer Cyber Security

Lecture 04 Block Ciphers, Modes of Operation

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Outline

Block Ciphers

DES and AES

Modes of Operation

Reading Assignment

- ▶ This lecture: UC 3, 4 except 4.3, 5.1 – 5.1.5
- ▶ Next lecture (Wed. 9/2): Go Introduction
 - ▶ Please install VSCode and Go following the instructions on:
<https://docs.microsoft.com/en-us/azure/developer/go/configure-visual-studio-code>

Outline

Block Ciphers

DES and AES

Modes of Operation

Overview

- ▶ Substitution cipher → OTP (unconditional security) → Stream ciphers (CSPRNG)
- ▶ How about cryptanalysis based on statistics?
- ▶ Simple substitution cipher maps letters to letters.
 - ▶ If there is only 26 letters, collecting a few thousands letters (e.g. allow each letter to appear 100 times on average) of ciphertext will reveal substantial amount of statistics.
- ▶ For plaintext and ciphertext as bytes, need a few tens of thousand of bytes so each byte appear 100 times on average.
- ▶ What about substitution on larger blocks of bits?
 - ▶ E.g. 64-bit blocks: every block appears once on average in $2^{64} * 8$ bytes – seems longer than any practical message.
 - ▶ Need to study more to be a secure cipher.

Block Ciphers

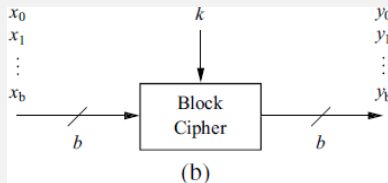


Fig.2 (Paar and Pelzl)

- ▶ Shared secret key k .
- ▶ Plaintext x as bit blocks of fixed size.
- ▶ Each block is encrypted via a block cipher and then concatenated into the ciphertext y .

Discussions

- ▶ We first focus on block encryption and decryption, i.e. both x and y are fixed-length bit strings.
 - ▶ Popular block lengths in bit: 64, 128, 256,
- ▶ A substitution cipher with 64-bit blocks need $(2^{64})!$ keys.
 - ▶ Generate random permutations if keys are chosen uniformly.
 - ▶ But not practical to store or transmit such keys.
- ▶ A block cipher only supports a subset of the permutations.
 - ▶ Still able to resist brute-force if the key space is large enough.
 - ▶ Key space depends on key sizes: 64-bit, 128-bit,
- ▶ With modern cryptanalysis, a strong block cipher should resist known-plaintext attack and beyond.
 - ▶ E.g. allow attackers to have access to $e_k()$ or $d_k()$ (without knowing k directly).

Outline

Block Ciphers

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History of Data Encryption Standard (DES)

- ▶ 1972: NBS (now NIST) request for proposals for a standardized cipher in the USA
 - ▶ Motivated by demands for encryption in commercial applications.
 - ▶ Before this, cryptography and cryptanalysis are considered so crucial for national security that it had to be kept secret.
- ▶ 1974: proposal from IBM received
- ▶ 1977: NBS release Data Encryption Standard (FIPS PUB 46)
 - ▶ IBM cipher modified by NSA.
- ▶ 1990's: key space too small (2^{56}) to resist brute-force attack
 - ▶ Moore's law: computers become much more powerful
 - ▶ Triple DES proposed as a remedy
- ▶ 2001: NIST publish Advanced Encryption Standard (AES)
 - ▶ This is what you should use instead of DES as of now.

History of Advanced Encryption Standard (AES)

- ▶ 1997: NIST call for proposals
 - ▶ 128-bit block with 128, 192, and 256 bits keys
 - ▶ Efficiency in software and hardware
 - ▶ Open selection process
- ▶ 1998: 15 candidate algorithms, from several countries
- ▶ 1999: 5 finalist algorithms
 - ▶ Mars, RC6, Rijndael, Serpent, Twofish
- ▶ 2000: Rijndael announced as the winner
- ▶ 2001: Advanced Encryption Standard (AES) (FIPS PUB 197)
- ▶ 2003: NSA announced that it allows AES to encrypt classified documents up to the level SECRET, and up to the TOP SECRET level for 192 or 256-bit keys.

AES Encryption

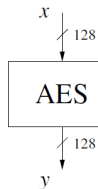


Fig. 4.1 AES input/output parameters

key lengths	# rounds = n_r
128 bit	10
192 bit	12
256 bit	14

Table 4.1

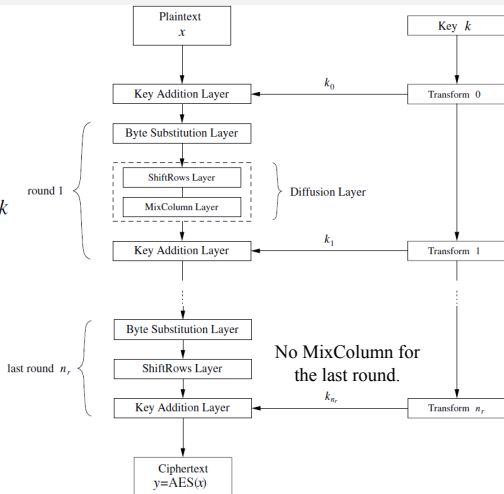


Fig. 4.2 AES encryption block diagram

(Paar and Pelzl)

- Round keys are always 128 bits.

AES Decryption

- ▶ Need to invert all layers.
 - ▶ Need extra resource though the basic structure is similar.
- ▶ Key schedule remains the same.
 - ▶ The order to apply subkeys are reversed.

AES Implementations

- ▶ A lot of literatures as references.
- ▶ Hardware
 - ▶ ASIC or FPGA
 - ▶ Optimized for throughput, e.g. for 400Gb/s and beyond networking, or power/area, e.g. for IoT devices.
- ▶ Software
 - ▶ Purely software: table lookup
 - ▶ Hardware acceleration: e.g. AES-NI for x86 CPUs
 - ▶ Don't implement it by yourself, use a library for correctness, security, and performance.

Outline

Block Ciphers

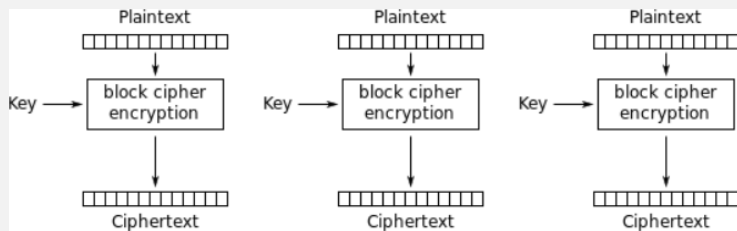
DES and AES

Modes of Operation

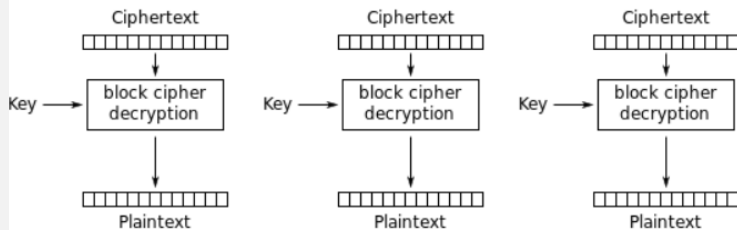
Should we apply AES as it is directly to messages?

- ▶ What if the message is longer than 128 bits?
- ▶ What if the message is not exactly 128 bits?
- ▶ Any other concerns?
- ▶ What about other block ciphers?

Electronic Code Book (ECB)



Electronic Codebook (ECB) mode encryption

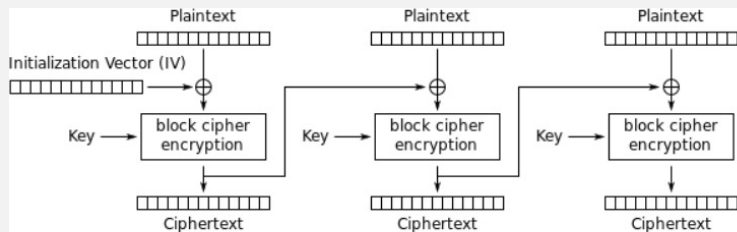


Electronic Codebook (ECB) mode decryption

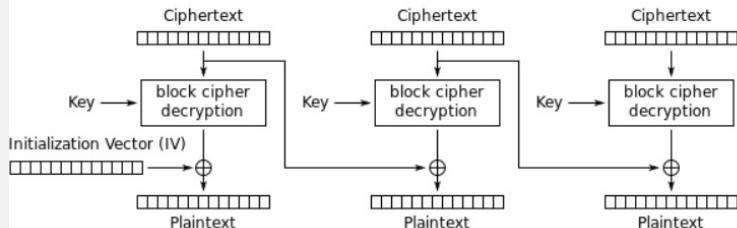
(Wikipedia)

- ▶ A substitution cipher based on a block cipher like AES.
- ▶ Padding: when message size is not multiples of block size
 - ▶ Alice appends additional bits that Bob will identify.
 - ▶ E.g. 1 followed by necessary number of 0's.
- ▶ Oscar the passive adversary
 - ▶ Known-plaintext attack using padding.
 - ▶ Traffic analysis possible since same plaintext blocks always encrypts to same ciphertext blocks.
- ▶ Can be parallelized as long as the message is available.

Cipher Block Chaining (CBC)



Cipher Block Chaining (CBC) mode encryption

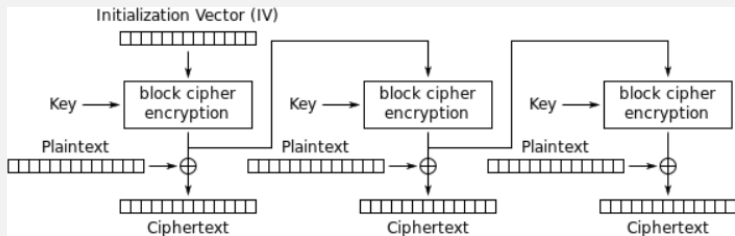


Cipher Block Chaining (CBC) mode decryption

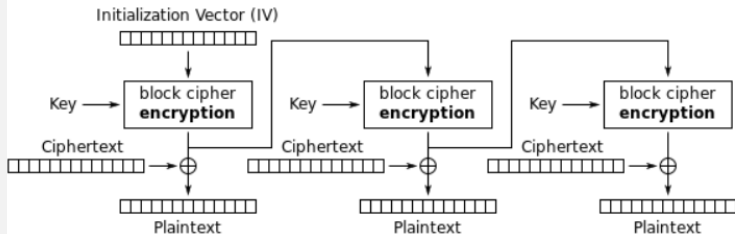
(Wikipedia)

- ▶ “Randomize” plaintext blocks
 - ▶ Use previous ciphertext blocks.
 - ▶ Use an initialization vector (IV) for the first plaintext block.
- ▶ Choice of IV
 - ▶ Probabilistic encryption: different IVs results in different ciphertexts even if the plaintext and the key are the same.
 - ▶ A.k.a nonce – a number used only once.
 - ▶ Usually randomly chosen and transmitted before ciphertext.
 - ▶ Oscar will see it.
 - ▶ If that’s a concern, Alice could just encrypt IV.
- ▶ Only decryption can be parallelized.

Output Feedback (OFB)



Output Feedback (OFB) mode encryption

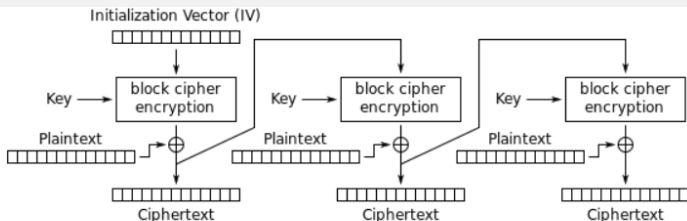


Output Feedback (OFB) mode decryption

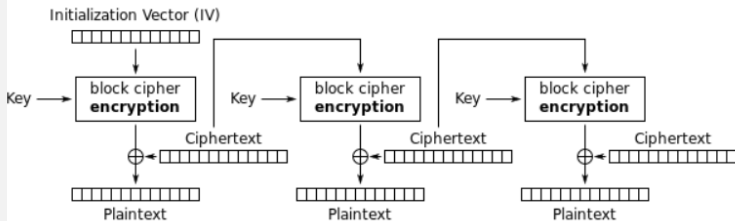
(Wikipedia)

- ▶ A stream cipher based on a block cipher.
 - ▶ Random IV guarantees probabilistic encryption.
 - ▶ Resist known-plaintext attack as long as the block cipher can resist known-plaintext attack.
- ▶ Only need encryption from the block cipher.
 - ▶ No need to implement decryption – save hardware resource.
- ▶ Cannot be parallelized.
 - ▶ Key stream can be precomputed as long as storage permits.

Cipher Feedback (CFB)



Cipher Feedback (CFB) mode encryption

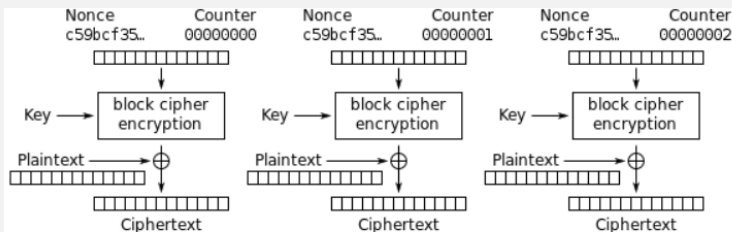


Cipher Feedback (CFB) mode decryption

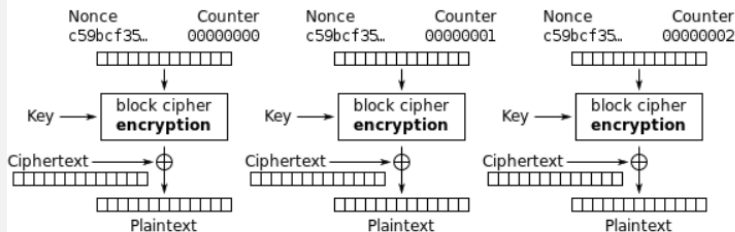
(Wikipedia)

- ▶ An asynchronous stream cipher as the key stream depends on both key and previous ciphertext (and plaintext).
 - ▶ Otherwise very similar to OFB.
- ▶ Only need encryption from the block cipher.
- ▶ Decryption can be parallelized.

Counter Mode (CTR)



Counter (CTR) mode encryption



Counter (CTR) mode decryption

(Wikipedia)

- ▶ A stream cipher that can be fully parallelized.
 - ▶ Resist known-plaintext attack as long as the block cipher can resist known-plaintext attack.
- ▶ Only need encryption as OFB and CFB.
- ▶ There is a limitation on message size for a given IV.
 - ▶ OFB also has limitation on message size, although it should be much longer.

Active Adversaries and Integrity

- ▶ We introduce passive adversaries to address confidentiality.
- ▶ For integrity, we could address it by active adversaries.
 - ▶ They can modify or even insert messages.
 - ▶ E.g. reorder/substitute/modify/create blocks.
- ▶ With the ability to manipulate ciphertext, active adversaries could even
 - ▶ Break confidentiality by side-channel attack.
 - ▶ Break higher level protocols by replay attack.
- ▶ None of the modes of operation can guarantee integrity.
 - ▶ **No matter how secure the underlying block cipher is.**
 - ▶ E.g. if reordering and substitution attacks are applied to ECB, all blocks will decrypt correctly but may mean things completely different when combined together.

Summary

- ▶ Block ciphers
- ▶ DES and AES
- ▶ Modes of operation